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(57) Abstract:

The given application provides a Fogg based Explainable AI and detection AI (YOLOV8) framework for plant disease. This system integrates a mobile edge layer, a fog computation layer, and a cloud service that can assure efficiency, reliability, accuracy and less latency. The mobile edge part of the framework performs preprocessing, image resizing. Image detection and Explainable AI is done through YOLOV8 and Grad-CAM respectively. YOLOV8, unlike its previous versions, has lesser complexity and better performance .This construction aims to provide an app that is more user friendly and comfortable for farmers and rural environment.

Fog assisted explainable AI for plant disease detection

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Description:

Background of the Invention: This system was invented for farmers to detect diseases in crops, which can lead to timely intervention and help them take the necessary steps. The use of fog based AI framework makes this system more convenient to use even in places with low bandwidth and connectivity issues, compared to other apps. It is also very user friendly including regional languages not just English to make it easier to comprehend.

Summary of the Invention: This invention offers a 3 layered application that balances between mobile edge, fog layer, and cloud services layer. The application has a mobile edge layer, that deals with capturing the image as input and performing light weight preprocessing, contrast enhancement to make the detection process more convenient efficient. The backend server uses YOLOv8 as detection module.

The Fog computing layer acts as an intermediate framework that receives the preprocessed image and uses it to run detection and explainable AI model to help us generate the disease detection results. It also helps with reducing mobile battery consumption.

The cloud service layer is responsible to perform all the heavy tasks that is impossible to be performed on the mobile phone such as data acquisition i.e. preparing huge amounts of data, and the optimization of model where AI model is trained, refined and optimized to their best accuracy. It also serves as a storage hub for the entire project.

FIG. 1. Overall architecture of the Fog assisted explainable AI for plants disease

FIG. 2. Mobile edge layer

FIG. 3 Fog based layer framework

FIG. 4. Cloud based model training and optimization

Description of Preferred Embodiment:

With reference to the drawings and in particular to FIGS. 1 and 2, FIG. 1 shows the modules of the system while also highlighting the three-tier architecture; FIG. 2 is an overall view on all the steps followed by the system to analyze a plant, detect disease and save the results.

In FIG 1 modules of the system the three-tier physical architecture of the system is shown which the physical infrastructure of the process that occurs is. The mobile Edge layer is the bottom-most layer that handles image acquisition and secure transmission, communication protocols. The fog based intelligent layer handles explainable AI module working and integration into the system while the cloud- based data Processing module does the heavy work of deep learning model training and nodal optimization which is how models are trained and improved over time.

In FIG 2 the overall process is shown step-by-step on how it works. The image is acquired from the user input preprocessed before model is run on it and the detection model is run on it to predict the disease. This forms the overall Image acquisition and Local Screening Module. The communication and Data Offloading Module handles the connectivity between all these steps, packing od data which are results derived from the model and ensures secure transmission of these data packets.

In FIG 3 the transition of how data becomes expert decisions is visualized. The data is cleaned, annotated, validated and preprocessed before running the detection model on it. The focus is to not just make the detection but also make sure the output is easy to comprehend for the farmers. The visual Explanation module focuses on visualization of the output to make the output easy to understand. This includes the Explainable AI component. The decision support system helps with providing actionable advice to the farmers.

In FIG 4 it explains the development progress of the entire project. It can be divided into 3 stages which are the data acquisition and preparation, mobile development and optimization and

deployment stages. The data acquisition and preparation stage involves collection of the required amount of data in the required format, cleaning, validating and preprocessing of it to prepare them for model training stage. The mobile Development and optimization stage is the core part of the project where model is trained,

Optimized and then evaluated. The deployment stage is when the finished, optimized model is integrated into the KrishiLens android app and completes the entire project so that farmers can start using it for their benefit.

1. Distributed three-tier architecture consisting of a mobile-edge layer, fog assisted framework and a cloud-based model where the mobile-edge layer handles image input and basic preprocessing, fog assisted framework acts as the middleman and integrates explainability, and cloud-based model handles the heavy model training work.
2. The fog layer as claimed in claim 1, assists with visual evidence generation for the diseases detected.
3. The fog layer as claimed in claim 1, assigned with explainable AI functions helps with reduction in mobile battery consumption and more efficient use.
4. The mobile edges layer as claimed in claim 1, preprocesses the image given and the backended server uses YOLOv8 model for detection
5. The user interface displays the selected image, the name of the disease detected (e.g. “Tomato leaf blight”) and a confidence score in percentage.
6. The YOLOv8 detection model shows boxes to identify various diseases simultaneously in a single picture. So, it helps with multiple disease prediction in the same image rather than classifying or mapping each image to a single disease